

■ Database Availability

KEY WORDS

- Block Corruption
- Data pump
- dbms_metadata

<<Block Corruption>>

```
SYS@orcl> drop table hr.emp purge;
```

```
SYS@orcl> create table hr.emp tablespace users as select * from hr.employees;
```

Table created.

```
SYS@orcl> select file_id, block_id from dba_extents where segment_name = 'EMP' and owner = 'HR';
```

FILE_ID	BLOCK_ID
7	360

```
SYS@orcl> select f.tablespace_name, f.file_name, count(*) from dba_extents e, dba_data_files f where e.file_id = f.file_id and e.segment_name = 'EMP' and e.owner = 'HR' group by f.tablespace_name, f.file_name;
```

TABLESPACE_NAME	FILE_NAME	COUNT(*)
USERS	/u01/app/oracle/oradata/ORCL/users01.dbf	1

```
SYS@orcl> select employee_id, rowid, dbms_rowid.rowid_block_number(rowid) from hr.emp;
```

EMPLOYEE_ID	ROWID	DBMS_ROWID.ROWID_BLOCK_NUMBER(ROWID)
100	AAASA8AAHAAAFrAAA	363
101	AAASA8AAHAAAFrAAB	363
102	AAASA8AAHAAAFrAAC	363
103	AAASA8AAHAAAFrAAD	363
104	AAASA8AAHAAAFrAAE	363
105	AAASA8AAHAAAFrAAF	363
106	AAASA8AAHAAAFrAAG	363
205	AAASA8AAHAAAFsAAH	364
206	AAASA8AAHAAAFsAAI	364

107 rows selected.

- Block 훼손

```
SYS@orcl> !dd if=/dev/zero of=/u01/app/oracle/oradata/ORCL/users01.dbf bs=8192 seek=364 count=2
```

conv=notrunc

2+0 records in

2+0 records out

16384 bytes (16 kB) copied, 0.000152238 s, 108 MB/s

SYS@orcl> alter system flush buffer_cache;

System altered.

SYS@orcl> /

System altered.

SYS@orcl> select count(*) from hr.emp;

select count(*) from hr.emp

*

ERROR at line 1:

ORA-01578: ORACLE data block corrupted (file # 7, block # 364)

ORA-01110: data file 7: '/u01/app/oracle/oradata/ORCL/users01.dbf'

-- 훼손 Block 찾기

SYS@orcl> !dbv userid=system/oracle file=/u01/app/oracle/oradata/ORCL/users01.dbf

DBVERIFY - Verification starting : FILE = /u01/app/oracle/oradata/ORCL/users01.dbf

Page 364 is marked corrupt

Corrupt block relative dba: 0x01c0016c (file 7, block 364)

Completely zero block found during dbv:

Page 365 is marked corrupt

Corrupt block relative dba: 0x01c0016d (file 7, block 365)

Completely zero block found during dbv:

Page 395 is marked corrupt

Corrupt block relative dba: 0x01c0018b (file 7, block 395)

Completely zero block found during dbv:

Page 396 is marked corrupt

Corrupt block relative dba: 0x01c0018c (file 7, block 396)

Completely zero block found during dbv:

DBVERIFY - Verification complete

Total Pages Examined : 640
 Total Pages Processed (Data) : 67
 Total Pages Failing (Data) : 0
 Total Pages Processed (Index): 15
 Total Pages Failing (Index): 0
 Total Pages Processed (Other): 441
 Total Pages Processed (Seg) : 30
 Total Pages Failing (Seg) : 0
 Total Pages Empty : 83
 Total Pages Marked Corrupt : 4
 Total Pages Influx : 0
 Total Pages Encrypted : 0
 Highest block SCN : 5005984 (0.5005984)

```

SYS@orcl> select t.ts#, s.header_file, s.header_block
from v$tablespace t, dba_segments s
where t.name = s.tablespace_name
and s.segment_name = 'EMP'
and s.owner = 'HR';
  
```

TS#	HEADER_FILE	HEADER_BLOCK
4	7	362

```

SYS@orcl> !dbv userid=system/oracle segment_id=4.7.362
  
```

DBVERIFY: Release 19.0.0.0.0 - Production on Thu May 22 21:05:44 2025

DBVERIFY - Verification starting : SEGMENT_ID = 4.7.362

Page 364 is marked corrupt

Corrupt block relative dba: 0x01c0016c (file 7, block 364)

Completely zero block found during dbv:

Page 365 is marked corrupt

Corrupt block relative dba: 0x01c0016d (file 7, block 365)

Completely zero block found during dbv:

DBVERIFY - Verification complete

Total Pages Examined : 8
 Total Pages Processed (Data) : 1
 Total Pages Failing (Data) : 0
 Total Pages Processed (Index): 0
 Total Pages Failing (Index): 0
 Total Pages Processed (Other): 2
 Total Pages Processed (Seg) : 1
 Total Pages Failing (Seg) : 0
 Total Pages Empty : 2
 Total Pages Marked Corrupt : 2
 Total Pages Influx : 0
 Total Pages Encrypted : 0
 Highest block SCN : 5005984 (0.5005984)

RMAN> validate datafile 7;

Starting validate at 22-MAY-25

using channel ORA_DISK_1

channel ORA_DISK_1: starting validation of datafile

channel ORA_DISK_1: specifying datafile(s) for validation

input datafile file number=00007 name=/u01/app/oracle/oradata/ORCL/users01.dbf

channel ORA_DISK_1: validation complete, elapsed time: 00:00:01

List of Datafiles

=====

File Status Marked Corrupt Empty Blocks Blocks Examined High SCN

File	Status	Marked Corrupt	Empty Blocks	Blocks Examined	High SCN
7	FAILED	1	83	642	5005984

File Name: /u01/app/oracle/oradata/ORCL/users01.dbf

Block Type Blocks Failing Blocks Processed

Block Type	Blocks Failing	Blocks Processed
Data	0	67
Index	0	15
Other	3	475

validate found one or more corrupt blocks

See trace file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl_ora_19058.trc for details

Finished validate at 22-MAY-25

RMAN> list failure;

Database Role: PRIMARY

List of Database Failures

=====

Failure ID	Priority	Status	Time Detected	Summary
8262	HIGH	OPEN	22-MAY-25	Datafile 7: '/u01/app/oracle/oradata/ORCL/users01.dbf' contains one or more corrupt blocks

RMAN> list failure 8262 detail;

Database Role: PRIMARY

List of Database Failures

=====

Failure ID	Priority	Status	Time Detected	Summary
8262	HIGH	OPEN	22-MAY-25	Datafile 7: '/u01/app/oracle/oradata/ORCL/users01.dbf' contains one or more corrupt blocks

Impact: Some objects in tablespace USERS might be unavailable

List of child failures for parent failure ID 8262

Failure ID	Priority	Status	Time Detected	Summary
8302	HIGH	OPEN	22-MAY-25	Block 396 in datafile 7: '/u01/app/oracle/oradata/ORCL/users01.dbf' is media corrupt

Impact: Object owned by might be unavailable

8296	HIGH	OPEN	22-MAY-25	Block 395 in datafile 7: '/u01/app/oracle/oradata/ORCL/users01.dbf' is media corrupt
------	------	------	-----------	--

Impact: Object owned by might be unavailable

8271	HIGH	OPEN	22-MAY-25	Block 365 in datafile 7: '/u01/app/oracle/oradata/ORCL/users01.dbf' is media corrupt
------	------	------	-----------	--

Impact: Object EMP owned by HR might be unavailable

8265	HIGH	OPEN	22-MAY-25	Block 364 in datafile 7: '/u01/app/oracle/oradata/ORCL/users01.dbf' is media corrupt
------	------	------	-----------	--

Impact: Object EMP owned by HR might be unavailable

RMAN> advise failure;

Database Role: PRIMARY

List of Database Failures

=====

Failure ID Priority Status Time Detected Summary

Failure ID	Priority	Status	Time Detected	Summary
8262	HIGH	OPEN	22-MAY-25	Datafile 7: '/u01/app/oracle/oradata/ORCL/users01.dbf' contains one or more corrupt blocks

Impact: Some objects in tablespace USERS might be unavailable

List of child failures for parent failure ID 8262

Failure ID Priority Status Time Detected Summary

Failure ID	Priority	Status	Time Detected	Summary
8302	HIGH	OPEN	22-MAY-25	Block 396 in datafile 7: '/u01/app/oracle/oradata/ORCL/users01.dbf' is media corrupt

Impact: Object owned by might be unavailable

Failure ID	Priority	Status	Time Detected	Summary
8296	HIGH	OPEN	22-MAY-25	Block 395 in datafile 7: '/u01/app/oracle/oradata/ORCL/users01.dbf' is media corrupt

Impact: Object owned by might be unavailable

Failure ID	Priority	Status	Time Detected	Summary
8271	HIGH	OPEN	22-MAY-25	Block 365 in datafile 7: '/u01/app/oracle/oradata/ORCL/users01.dbf' is media corrupt

Impact: Object EMP owned by HR might be unavailable

Failure ID	Priority	Status	Time Detected	Summary
8265	HIGH	OPEN	22-MAY-25	Block 364 in datafile 7: '/u01/app/oracle/oradata/ORCL/users01.dbf' is media corrupt

Impact: Object EMP owned by HR might be unavailable

analyzing automatic repair options; this may take some time

using channel ORA_DISK_1

analyzing automatic repair options complete

Mandatory Manual Actions

=====

no manual actions available

Optional Manual Actions

=====

no manual actions available

Automated Repair Options

=====

Option Repair Description

Option	Repair Description
1	Recover multiple corrupt blocks in datafile 7

Strategy: The repair includes complete media recovery with no data loss

Repair script: /u01/app/oracle/diag/rdbms/orcl/orcl/hm/reco_2034484387.hm

RMAN> repair failure preview;

Strategy: The repair includes complete media recovery with no data loss

Repair script: /u01/app/oracle/diag/rdbms/orcl/orcl/hm/reco_2034484387.hm

contents of repair script:

```
# block media recovery for multiple blocks
recover datafile 7 block 364 to 365, 395 to 396;
```

RMAN> repair failure;

Strategy: The repair includes complete media recovery with no data loss

Repair script: /u01/app/oracle/diag/rdbms/orcl/orcl/hm/reco_2034484387.hm

contents of repair script:

```
# block media recovery for multiple blocks
recover datafile 7 block 364 to 365, 395 to 396;
```

Do you really want to execute the above repair (enter YES or NO)? yes
executing repair script

Starting recover at 22-MAY-25
using channel ORA_DISK_1

channel ORA_DISK_1: restoring block(s) from datafile copy
/home/oracle/backup/FRA/ORCL/datafile/o1_mf_users_n2s7b9wt_.dbf

starting media recovery
media recovery complete, elapsed time: 00:00:03

Finished recover at 22-MAY-25
repair failure complete

RMAN> list failure;

Database Role: PRIMARY

no failures found that match specification

RMAN> select count(*) from hr.emp;

COUNT(*)

107

<< Data pump >>

```
SYS@orcl> !mkdir /home/oracle/datapump
```

```
SYS@orcl> create directory dirpump as '/home/oracle/datapump';
```

Directory created.

```
SYS@orcl> select * from dba_directories where directory_name = 'DIRPUMP';
```

OWNER	DIRECTORY_NAME	DIRECTORY_PATH	ORIGIN_CON_ID
SYS	DIRPUMP	/home/oracle/datapump	0

```
SYS@orcl> Grant read, write on directory dirpump to hr;
```

Grant succeeded.

```
SYS@orcl> select * from dba_tab_privs where grantee = 'HR';
```

- Schema level Export

```
SYS@orcl> !expdp system/oracle schemas=hr directory=dirpump dumpfile=hr_schema.dmp
```

```
SYS@orcl> drop table hr.emp purge;
```

Table dropped.

```
SYS@orcl> select count(*) from hr.emp;
```

```
select count(*) from hr.emp
```

*

ERROR at line 1:

ORA-00942: table or view does not exist

- Table level Import

```
SYS@orcl> ! impdp system/oracle directory=dirpump dumpfile=hr_schema.dmp tables=hr.emp
```

```
SYS@orcl> select count(*) from hr.emp;
```

COUNT(*)

107

```
SYS@orcl> select constraint_name, constraint_type, search_condition, status, index_name from dba_constraints
where table_name = 'EMP' and owner = 'HR';
```

-Remap Operation(Table level) Export

```
SYS@orcl> ! expdp system/oracle schemas=hr directory=dirpump dumpfile=hr_table.dmp include=table
```

```
SYS@orcl> select f.tablespace_name, f.file_name from dba_extents e, dba_data_files f where e.file_id = f.file_id and  
e.segment_name='EMP' and e.owner='HR';
```

TABLESPACE_NAME	FILE_NAME

USERS	/u01/app/oracle/oradata/ORCL/users01.dbf

```
SYS@orcl> select table_name, tablespace_name from dba_tables where owner = 'HR';
```

TABLE_NAME	TABLESPACE_NAME

EMP_TEMP	USERS
REGIONS	SYSAUX
COUNTRIES	
LOCATIONS	SYSAUX
DEPARTMENTS	SYSAUX
JOBS	SYSAUX
EMPLOYEES	SYSAUX
JOB_HISTORY	SYSAUX
TEST_DML	USERS
LOC1	USERS
EMP2	SHTBS
DEPT1	OETBS
EMP3	OETBS
EMP	USERS

14 rows selected.

```
SYS@orcl> create user kkt identified by kkt
```

- 2 default tablespace users
- 3 temporary tablespace temp
- 4 quota unlimited on users;

User created.

```
SYS@orcl> grant connect, resource to kkt;
```

Grant succeeded.

- Remap Tablespace Import

```
SYS@orcl> ! impdp system/oracle directory=dirpump dumpfile=hr_table.dmp tables=hr.jobs  
remap_schema='hr':'kkt' remap_tablespace='sysaux':'users'
```

Import: Release 19.0.0.0.0 - Production on Thu May 22 22:37:45 2025

Version 19.3.0.0.0

Connected to: Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production

Master table "SYSTEM"."SYS_IMPORT_TABLE_01" successfully loaded/unloaded

Starting "SYSTEM"."SYS_IMPORT_TABLE_01": system/***** directory=dirpump dumpfile=hr_table.dmp

tables=hr.jobs remap_schema=hr:kkt remap_tablespace=sysaux:users

Processing object type SCHEMA_EXPORT/TABLE/TABLE

Processing object type SCHEMA_EXPORT/TABLE/TABLE_DATA

. . imported "KKT"."JOBS" 7.109 KB 19 rows

Processing object type SCHEMA_EXPORT/TABLE/COMMENT

Processing object type SCHEMA_EXPORT/TABLE/INDEX/INDEX

Processing object type SCHEMA_EXPORT/TABLE/CONSTRAINT/CONSTRAINT

Processing object type SCHEMA_EXPORT/TABLE/INDEX/STATISTICS/INDEX_STATISTICS

Processing object type SCHEMA_EXPORT/TABLE/STATISTICS/TABLE_STATISTICS

Job "SYSTEM"."SYS_IMPORT_TABLE_01" successfully completed at Thu May 22 22:37:54 2025 elapsed 0 00:00:06

SYS@orcl> conn kkt/kkt

Connected.

KKT@orcl> select table_name, tablespace_name from user_tables;

TABLE_NAME	TABLESPACE_NAME
JOBS	USERS

JOBS USERS

KKT@orcl> select count(*) from jobs;

COUNT(*)

19

- Remap Schema

KKT@orcl> !impdp system/oracle directory=dirpump dumpfile=hr_table.dmp

tables=hr.emp,hr.employees,hr.departments,hr.locations remap_schema='hr':kkt' exclude=trigger,ref_constraint

- 특정 Table Export/Import

SYS@orcl> !expdp system/oracle directory=dirpump dumpfile=hr_emp_dept_job.dmp

tables=hr.employees,hr.departments,hr.job_history,kkt.emp

Connected to: Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production

Starting "SYSTEM"."SYS_EXPORT_TABLE_01": system/***** directory=dirpump dumpfile=hr_emp_dept_job.dmp

tables=hr.employees,hr.departments,hr.job_history,kkt.emp

Processing object type TABLE_EXPORT/TABLE/TABLE_DATA

Processing object type TABLE_EXPORT/TABLE/INDEX/STATISTICS/INDEX_STATISTICS

Processing object type TABLE_EXPORT/TABLE/STATISTICS/TABLE_STATISTICS

Processing object type TABLE_EXPORT/TABLE/STATISTICS/MARKER

Processing object type TABLE_EXPORT/TABLE/TABLE

Processing object type TABLE_EXPORT/TABLE/COMMENT

Processing object type TABLE_EXPORT/TABLE/INDEX/INDEX

Processing object type TABLE_EXPORT/TABLE/CONSTRAINT/CONSTRAINT

Processing object type TABLE_EXPORT/TABLE/CONSTRAINT/REF_CONSTRAINT

Processing object type TABLE_EXPORT/TABLE/TRIGGER

```
.. exported "HR"."EMPLOYEES"                17.07 KB      107 rows
.. exported "HR"."JOB_HISTORY"              7.195 KB       10 rows
.. exported "HR"."DEPARTMENTS"             7.125 KB       27 rows
.. exported "KKT"."EMP"                    17.07 KB      107 rows
```

Master table "SYSTEM"."SYS_EXPORT_TABLE_01" successfully loaded/unloaded

Dump file set for SYSTEM.SYS_EXPORT_TABLE_01 is:

/home/oracle/datapump/hr_emp_dept_job.dmp

Job "SYSTEM"."SYS_EXPORT_TABLE_01" successfully completed at Thu May 22 22:50:49 2025 elapsed 0 00:00:24

SYS@orcl> select salary from kkt.emp where employee_id=100;

SALARY

30000

SYS@orcl> update kkt.emp set salary=5000 where employee_id=100;

1 row updated.

SYS@orcl> select salary from kkt.emp where employee_id=100;

SALARY

5000

SYS@orcl> commit;

Commit complete.

SYS@orcl> drop table kkt.emp purge;

Table dropped.

```
SYS@orcl> !impdp system/oracle directory=dirpump dumpfile=hr_emp_dept_job.dmp tables=kkt.emp
```

Connected to: Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production

Master table "SYSTEM"."SYS_IMPORT_TABLE_01" successfully loaded/unloaded

Starting "SYSTEM"."SYS_IMPORT_TABLE_01": system/***** directory=dirpump dumpfile=hr_emp_dept_job.dmp
tables=kkt.emp

Processing object type TABLE_EXPORT/TABLE/TABLE

Processing object type TABLE_EXPORT/TABLE/TABLE_DATA

.. imported "KKT"."EMP" 17.07 KB 107 rows

Processing object type TABLE_EXPORT/TABLE/STATISTICS/TABLE_STATISTICS

Processing object type TABLE_EXPORT/TABLE/STATISTICS/MARKER

Job "SYSTEM"."SYS_IMPORT_TABLE_01" successfully completed at Thu May 22 23:07:03 2025 elapsed 0 00:00:19

```
SYS@orcl> select salary from kkt.emp where employee_id=100;
```

SALARY

30000

- Data만 Import

```
SYS@orcl> truncate table hr.job_history;
```

Table truncated.

```
SYS@orcl> select count(*) from hr.job_history;
```

COUNT(*)

0

```
SYS@orcl> !impdp system/oracle directory=dirpump dumpfile=hr_emp_dept_job.dmp tables=hr.job_history  
content=data_only
```

Connected to: Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production

Master table "SYSTEM"."SYS_IMPORT_TABLE_01" successfully loaded/unloaded

Starting "SYSTEM"."SYS_IMPORT_TABLE_01": system/***** directory=dirpump dumpfile=hr_emp_dept_job.dmp
tables=hr.job_history content=data_only

Processing object type TABLE_EXPORT/TABLE/TABLE_DATA

.. imported "HR"."JOB_HISTORY" 7.195 KB 10 rows

Job "SYSTEM"."SYS_IMPORT_TABLE_01" successfully completed at Thu May 22 23:10:04 2025 elapsed 0 00:00:07

```
SYS@orcl> select count(*) from hr.job_history;
```

```
COUNT(*)
```

```
-----
```

```
10
```

- Table구조만 Import

```
SYS@orcl> drop table hr.job_history purge;
```

Table dropped.

```
SYS@orcl> select count(*) from hr.job_history;
```

```
select count(*) from hr.job_history
```

```
*
```

ERROR at line 1:

ORA-00942: table or view does not exist

```
SYS@orcl> ! impdp system/oracle directory=dirpump dumpfile=hr_emp_dept_job.dmp tables=hr.job_history  
content=metadata_only
```

Connected to: Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production

Master table "SYSTEM"."SYS_IMPORT_TABLE_01" successfully loaded/unloaded

Starting "SYSTEM"."SYS_IMPORT_TABLE_01": system/***** directory=dirpump dumpfile=hr_emp_dept_job.dmp
tables=hr.job_history content=metadata_only

Processing object type TABLE_EXPORT/TABLE/TABLE

Processing object type TABLE_EXPORT/TABLE/COMMENT

Processing object type TABLE_EXPORT/TABLE/INDEX/INDEX

Processing object type TABLE_EXPORT/TABLE/CONSTRAINT/CONSTRAINT

Processing object type TABLE_EXPORT/TABLE/INDEX/STATISTICS/INDEX_STATISTICS

Processing object type TABLE_EXPORT/TABLE/CONSTRAINT/REF_CONSTRAINT

Processing object type TABLE_EXPORT/TABLE/STATISTICS/TABLE_STATISTICS

Processing object type TABLE_EXPORT/TABLE/STATISTICS/MARKER

Job "SYSTEM"."SYS_IMPORT_TABLE_01" successfully completed at Thu May 22 23:13:16 2025 elapsed 0 00:00:20

```
SYS@orcl> select count(*) from hr.job_history;
```

```
COUNT(*)
```

```
-----
```

```
0
```

```
SYS@orcl> ! impdp system/oracle directory=dirpump dumpfile=hr_emp_dept_job.dmp tables=hr.job_history  
content=data_only
```

```
SYS@orcl> select count(*) from hr.job_history;  
COUNT(*)
```

```
-----  
10
```

- 동일 Table은 skip

```
SYS@orcl> ! impdp system/oracle directory=dirpump dumpfile=hr_emp_dept_job.dmp tables=hr.job_history  
table_exists_action=skip
```

- 동일한 Table이 있는 경우 데이터 추가

```
! impdp system/oracle directory=dirpump dumpfile=hr_emp_dept_job.dmp tables=hr.job_history  
table_exists_action=append <-- 에러 발생  
데이터 삭제 후 추가
```

```
! impdp system/oracle directory=dirpump dumpfile=hr_emp_dept_job.dmp tables=hr.job_history  
table_exists_action=truncate
```

- 동일한 Table이 있는 경우 삭제한 후 새로 생성하여 import

```
! impdp system/oracle directory=dirpump dumpfile=hr_emp_dept_job.dmp tables=hr.job_history  
table_exists_action=replace
```

- 조건절에 해당하는 데이터만 export

```
HR@orcl> ! expdp system/oracle directory=dirpump dumpfile=hr_50_stman.dmp tables=hr.employees  
query=hr.employees:W"where department_idW=50 and job_idW=W"ST_MANW"W"
```

```
HR@orcl> create table emp1 tablespace users
```

```
2 as select * from employees
```

```
3 where 1=2;
```

Table created.

```
HR@orcl> select * from emp1;
```

no rows selected

```
HR@orcl> ! impdp system/oracle directory=dirpump dumpfile=hr_50_stman.dmp remap_table=employees:emp1  
content=data_only
```

```
HR@orcl> select count(*) from emp1;  
COUNT(*)
```

```
-----  
5
```

-Expdp/Impdp using DB Link

- PROD DB 에서

```
SYSTEM@c1> create user itwill identified by itwill
```

```
2 default tablespace users
```

```
3 quota unlimited on users;
```

User created.

```
SYSTEM@c1> grant connect, resource to itwill;
```

Grant succeeded.

```
ITWILL@c1> conn itwill/itwill@c1
```

Connected.

```
ITWILL@c1> create table t1(id number);
```

Table created.

```
ITWILL@c1> insert into t1 values(1);
```

1 row created.

```
ITWILL@c1> commit;
```

Commit complete.

- orcl DB에서

```
SYS@orcl> create public database link prod_lk
```

```
2 connect to system identified by oracle
```

```
3 using 'c1';
```

```
[oracle@oel7v9r1 ~]$ expdp system/oracle directory=dirpump network_link=prod_lk schemas=itwill  
dumpfile=itwill.dmp
```

```
[oracle@oel7v9r1 ~]$ ! impdp system/oracle directory=dirpump schemas=itwill dumpfile=itwill.dmp  
remap_schema=itwill:hr
```

```
HR@orcl> select * from t1;
```

ID

1

<< dbms_metadata >>

- User 생성 질의 메타데이터 활용

```
SYS@orcl> select dbms_metadata.get_ddl('USER','HR') from dual;
```

```
DBMS_METADATA.GET_DDL('USER','HR')
```

```
-----
CREATE USER "HR" IDENTIFIED BY VALUES 'S:BAC3944055EA620
11F70F12C04E1245AA30
```

- User System 권한 확인

```
SYS@orcl> select * from dba_sys_privs where grantee = 'HR';
```

```
GR PRIVILEGE                                ADM COM INH
```

```
-----
HR CREATE DATABASE LINK                     NO NO NO
HR ALTER SESSION                           NO NO NO
HR CREATE VIEW                               NO NO NO
HR UNLIMITED TABLESPACE                   NO NO NO
HR CREATE SESSION                           NO NO NO
HR CREATE SEQUENCE                           NO NO NO
HR CREATE SYNONYM                           NO NO NO
```

7 rows selected.

- User System 권한 부여 질의 생성

```
SYS@orcl> select 'grant '||privilege||' to '||grantee||';' system_grant from dba_sys_privs where grantee = 'HR';
```

```
SYSTEM_GRANT
```

```
-----
grant CREATE DATABASE LINK to HR;
grant ALTER SESSION to HR;
grant CREATE VIEW to HR;
grant UNLIMITED TABLESPACE to HR;
grant CREATE SESSION to HR;
grant CREATE SEQUENCE to HR;
grant CREATE SYNONYM to HR;
```

7 rows selected.

- User System 권한 부여 질의 생성 메타데이터 활용

```
SYS@orcl> select dbms_metadata.get_granted_ddl('SYSTEM_GRANT','HR') from dual;
```

```
DBMS_METADATA.GET_GRANTED_DDL('SYSTEM_GRANT','HR')
```

```

GRANT CREATE SESSION TO "HR"
GRANT ALTER SESSION TO "HR"
GRANT UNLIMITED

```

- User System 권한 회수용 질의 생성

```
SYS@orcl> select 'revoke '||privilege||' from '||grantee||';' system_grant from dba_sys_privs where grantee = 'HR';
```

SYSTEM_GRANT

```

revoke CREATE DATABASE LINK from HR;
revoke ALTER SESSION from HR;
revoke CREATE VIEW from HR;
revoke UNLIMITED TABLESPACE from HR;
revoke CREATE SESSION from HR;
revoke CREATE SEQUENCE from HR;
revoke CREATE SYNONYM from HR;
7 rows selected.

```

- User Object 권한 부여용 질의 생성 메타데이터 활용

```
SYS@orcl> select dbms_metadata.get_granted_ddl('OBJECT_GRANT','HR') from dual;
```

DBMS_METADATA.GET_GRANTED_DDL('OBJECT_GRANT','HR')

```
GRANT EXECUTE ON "SYS"."DBMS_STATS" TO "HR"
```

- User의 Object 권한 확인

```
SYS@orcl> select * from dba_tab_privs where grantee = 'HR';
```

- User의 quota 값 확인

```
SYS@orcl> select * from dba_ts_quotas where username = 'HR';
```

TABLESPACE_NAME

USERNAME

```

BYTES  MAX_BYTES  BLOCKS MAX_BLOCKS DRO

```

SYSAUX

HR

1638400 -1 200 -1 NO

- User의 quota 적용된 질의문 생성 메타데이터 활용

```
SYS@orcl> select dbms_metadata.get_granted_ddl('TABLESPACE_QUOTA','HR') from dual;
```

```
DBMS_METADATA.GET_GRANTED_DDL('TABLESPACE_QUOTA','HR')
```

```
-----

DECLARE
TEMP_COUNT NUMBER;
SQLSTR VARCHAR2(200);
BEGIN
SQLSTR := 'A
```

```
SYS@orcl> alter user hr quota 200m on users;
```

- User의 quota값 변경된 내용 확인

```
SYS@orcl> select dbms_metadata.get_granted_ddl('TABLESPACE_QUOTA','HR') from dual;
```

```
DBMS_METADATA.GET_GRANTED_DDL('TABLESPACE_QUOTA','HR')
```

```
-----

DECLARE
TEMP_COUNT NUMBER;
SQLSTR VARCHAR2(200);
BEGIN
SQLSTR := 'A
```

- User quota 값 부여 질의

```
SYS@orcl> select 'alter user '||username||' quota '||decode(max_bytes,-1,'unlimited',max_bytes)||' on '||tablespace_name||';' quota from dba_ts_quotas where username = 'HR';
```

```
QUOTA
```

```
-----

alter user HR quota unlimited on SYSAUX;
alter user HR quota 209715200 on USERS;
```

- User 롤 부여 질의 생성 메타데이터

```
SYS@orcl> select dbms_metadata.get_granted_ddl('ROLE_GRANT','HR') from dual;
```

```
DBMS_METADATA.GET_GRANTED_DDL('ROLE_GRANT','HR')
```

```
GRANT "RESOURCE" TO "HR"
```

- User 롤 권한 확인

```
SYS@orcl> select * from dba_role_privs where grantee = 'HR';
```

```
GRANTEE
```

```
GRANTED_ROLE
```

```
ADM DEL DEF COM INH
```

```
HR
```

```
RESOURCE
```

```
NO NO YES NO NO
```